

Representation theory consequences

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G/E (conn'd) reductive gp / local field E , residue field $\overset{\mathbb{F}_q}{\underset{k}{\mathbb{Z}}} \leftarrow \mathcal{O}_E$

$\text{char } k = p > 0 \rightsquigarrow G(E)$ locally profinite

$\rightsquigarrow v\text{-stack } [* / G(E)]$ over $\text{Perf}_{\bar{k}} : (qcs) \text{ perf'd spaces over } \bar{k}$
 $= \text{Shv}_v(\text{Perf}_{\bar{k}}, \text{gpd})$

$$D[* / G(E), \Lambda] \simeq D(\text{Rep}^{\text{sm}}(G(E), \Lambda))$$

$D(-, \Lambda)$ t -functor formalism with coeff. $\Lambda \in \text{CRing } \mathbb{Z}[\frac{1}{p}]$

- Λ torsion : Scholze: étale coh. of diamond.

- Λ general: given by Bocklandt motive (better than D_{Lis} or D_{M})

Ex. $M \leftarrow P \rightarrow G$ parabolic P , Levi M

coh. smooth. $* / P(E)$
 $\swarrow \quad \searrow P \text{ proper}$
 $* / M(E) \quad * / G(E)$

$$I_P(E) = \text{Ind}_{P(E)}^{G(E)} := P_! q^* : D[* / M(E)] \rightarrow D[* / G(E)]$$

$$= P_* q^*$$

$$= R\gamma(q^!(-) \otimes \delta_P^{-1})$$

$$q^!(-) = q^*(-) \otimes \underline{q^!(\Lambda)}$$

$K_{* / P(E)}$ dualizing complex

$$\underline{q_! \left(\delta_P \otimes P^*(-) \right)} + I_{P(E)} \rightarrow q_* P^!$$

twist of $R_P = q_! P^*(-)$

Fact $= \delta_P$ invertible, i.e. character on $P(E)$, modular character

Fact I_p, R_p are exact (false for $\Lambda = \overline{\mathbb{F}_p}$)

I_p preserves admissible objects, R_p preserves compact objects

$D(*/\mathcal{A}(E))^\omega$ compact $\Leftrightarrow$ thick subcat. gen'd by $\{c\text{-ind}_K^G \Lambda\}_K$, K pro- p compact open

$D(*/\mathcal{A}(E))^{\text{Adm}}$ admissible $\Leftrightarrow V^K$ is perfect for any pro- p open cpt K

V adm. $\Leftrightarrow \text{Hom}(\text{cpt}, V)$ perfect, $\forall$ cpt

Second adjointness Up to a twist of S_p ,

right adjoint $(I_p)^R \simeq R_{\bar{p}}$, $\bar{p}$ opposite parabolic of p

$\Rightarrow (I_p)^R$ preserves colimit $\Leftrightarrow I_p$ preserves compact objects

$\Rightarrow R_p$ preserves adm. objects

$\downarrow$

V adm. $\text{Hom}(\text{cpt}, R_p V)$ perfect

$$\begin{array}{c} \parallel \\ \text{Hom}(I_p(\text{cpt}), V) \\ \uparrow \\ \text{cpt} \end{array}$$

Rem Geometric Version (Haman, Hansen, Scholze, Takeyama)

$\text{Bun}_G: S \in \text{Perf}_k \mapsto \{G\text{-torsors on } X_S\}$

normalized $\sqrt{q} \in \Lambda$

coh. smooth Bun_p

$\swarrow q$
 Bun_M

$\searrow p$ -representable
 Bun_G

$$\text{Eis}_p!(-) = p! \left(q^*(-) \otimes K_{\text{Bun}_p}^{1/2} \right)$$

$$\text{Eis}_p^*(-) = p^* \left(q^*(-) \otimes K_{\text{Bun}_p}^{1/2} \right)$$

$$\text{CT}_p^*(-) = q_* \left(p^!(-) \otimes K_{\text{Bun}_p}^{-1/2} \right)$$

$$\text{CT}_p!(-) = q! \left(p^*(-) \otimes K_{\text{Bun}_p}^{1/2} \right)$$

K_{Bun_p} dualizing complex on Bun_p

$$Eis p! \dashv CT_{P*}$$

$$CT_{P!} \dashv Eis p_*$$

Second adjointness $CT_{P!} \simeq CT_{\bar{P}*}$

$Eis p!$ preserves compact objects, CT_{P*} preserves LLA objects

(method of the proof: use second adjointness for $D(* / G(E))$).

Second adjointness $\xrightarrow[\text{Jacquet Lemma}]{\text{reduce to}}$ V smooth rep. $R_P(V^V) \simeq R_{\bar{P}}(V)^V$.

open up! K pro- $p < G(E)$, $K_M = K \cap M(E)$

want $(R_P(V^V))^{K_M} \simeq (R_{\bar{P}}(V))^{V, K_M}$

Stability K, P as above, V smooth rep of $G(E)$

One can choose K and $\lambda =$ totally positive central elts of $M(E)$

V is called K, P stable if $\exists$ constant (K, P, λ)

$$V^K = V_0^K \oplus V_{\lambda}^K \text{ s.t. } T_{\lambda} = \mathbb{1}_{K \times K} \curvearrowright V_{\lambda}^K \text{ invertibly}$$

$$\text{and } T_{\lambda}^{CK, P, \lambda}(V_0^K) = 0$$

such V is stable under $\ker$, coker , $\oplus$, π of uniform bounded (K, P, λ)

Our situation. $V^K \hookrightarrow T_{\lambda}$ V^K a $\Lambda[T_{\lambda}]$ -module

$$V^K \longrightarrow V^K \otimes_{\Lambda[T_{\lambda}]} \Lambda[T_{\lambda}^{\pm 1}] \xrightarrow[\text{(Jacquet lemma)}]{\simeq} R_P(V)^{K_M} = (V_u)^{K_M}$$

Lemma 33 B noetherian ring, M finite B -module, $\alpha \in \text{End}_B(M)$

if $M \otimes_{B[\alpha]} B[\alpha^{\pm 1}]$ is finitely gen'd $/B$, then $M = M[\alpha^{\infty}] \oplus M[\alpha^{\pm 1}]$
 $\parallel$
 $M[\alpha^{\pm 1}]$

if V^k finite module $/R$, if we can prove $R_p(V)^{k_M}$ also finite $/R$,

then we have stability

$$\begin{array}{ccc} V^k & \xrightarrow{\quad} & V^k \otimes \wedge[T_{\lambda}^{\pm 1}] = R_p(V)^{k_M} \\ & \nwarrow & \wedge[T_{\lambda}] \\ \parallel & & \\ V_0^k \oplus R_p(V)^{k_M} & = & V_0^k \oplus V_{\lambda}^k \end{array}$$

why stability $\Rightarrow$ Jacquet Lemma:

λ strict positive for p .

λ^{-1} strict positive for $\bar{p}$

$$V^{\vee k} = V_0^{\vee k} \oplus V^{\vee k}[T_{\lambda}^{\pm 1}]$$

$$V^k \stackrel{\text{we } \bar{p}}{=} V_0^k \oplus V^k[T_{\lambda^{-1}}^{\pm 1}]$$

$(V^{\vee})^k \times V^k \rightarrow \Delta$ perfect pairing, $T_{\lambda}, T_{\lambda^{-1}}$ adjoint to each other

$$\rightsquigarrow V^{\vee k}[T_{\lambda^{-1}}^{\pm 1}] \times V^k[T_{\lambda^{-1}}^{\pm 1}] \rightarrow \Delta$$

$$\begin{array}{ccc} \parallel & & \parallel \\ R_p(V^{\vee})^{k_M} \times R_{\bar{p}}(V)^{k_M} & \xrightarrow{\quad} & \Delta \end{array} \quad \text{perfect pairing}$$

Goal. Prove any smooth rep is K, P stable.

$\Uparrow$
 $\mathbb{Z}_\ell[\mathcal{H}(E)/H]_F$ is K, P -stable.
 $\uparrow$
 fix a depth

$V = \mathbb{Z}_\ell[G(E)/H]_r$ is admissible over $\mathfrak{Z}(G, \Lambda) = \pi_0 \text{End}(\text{id}_{D(* / \Lambda(E))})$

Want $R_p(V)$ is adm. over $\mathfrak{Z}(G, \Lambda)_{\mathbb{Z}_\ell}$ (over $\text{Exc}_{\mathbb{Z}_\ell}(W_{E/p}^\circ, \hat{G})_{r_1}$)

$\swarrow$ finitely gen'd $\xrightarrow{\text{main thm}}$ adm. over $\mathfrak{Z}(M, \mathbb{Z}_\ell)$
 $\xrightarrow{\text{finite}}$
 $\begin{cases} \mathfrak{Z}(G, \mathbb{Z}_\ell) \\ \text{Exc}_{\mathbb{Z}_\ell}(W_{E/p}^\circ, \hat{M})_{\text{red}} \end{cases}$

(Main thm: any smooth finitely gen'd rep. of $G(E)$ is $\mathfrak{Z}$ -finite)

Lemma $\mathfrak{Z}(G, \mathbb{Z}_\ell) \rightarrow \mathfrak{Z}(M, \mathbb{Z}_\ell)$ is finite

$\uparrow$
What is this map?

Hard to construct: find $\mathfrak{Z}(G, \mathcal{O}_\ell) \rightarrow \mathfrak{Z}(M, \mathcal{O}_\ell)$ (Bernstein block decomposition)

$\cup$ $\cup$
 $\mathfrak{Z}(G, \mathbb{Z}_\ell) \rightarrow \mathfrak{Z}(M, \mathbb{Z}_\ell)$
 $\uparrow$ $\uparrow$
 $\text{Exc}(W, \hat{G}) \rightarrow \text{Exc}(W, \hat{M})$

